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March 1, 1963
SUPERSEDING
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May 3, 1961

FEDERAL SPECIFICATION

NUT RUNNER, PNEUMATIC, PORTABLE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers one type of portable pneumatic nut runner for tightening and loosening bolts and nuts.

1.1.1 *Federal specification coverage.* Federal specifications do not cover all types of the commodity as indicated by the titles of the specifications, but include only those generally used by the Federal government.

1.2 Classification.

1.2.1 The nut runner covered by this specification shall be of one type and of the sizes indicated in table I and specified by the procuring agency (see 6.1).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-P-116—Preservation; Methods of.

MIL-R-196—Repair Parts for Internal Combustion Engines, Packaging of.

MIL-H-775—Hose, Rubber or Fabric (including Tubing); and Fittings, Nozzles and Strainers, Packaging of.

MIL-P-12829—Packaging of Pneumatic Hand Tools with Attachments and Accessories.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-130—Identification Marking of U. S. Military Property.

FSC 5130

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

National Bureau of Standards Handbook:

H28—Screw-Thread Standards for Federal Services.

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.).

American Standards Association Standard:

B5.38—Driving and Spindle Ends for Portable Air and Electric Tools.

(Copies may be obtained from the American Standards Association, Inc., 70 East 45th Street, New York 17, N. Y.)

3. REQUIREMENTS

3.1 **Illustrations.** The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of tools which are otherwise in accordance with the requirements of this specification.

3.2 **Qualification.** Portable pneumatic nut runners furnished under this specification shall be a product which has been tested, and has passed the qualification tests specified herein, and has been listed on, or approved for, listing on the applicable qualified products list.

3.3 **Material.** Material used in the construction of portable pneumatic nut runners shall be of a good commercial quality suitable for the intended purpose. All materials used shall be free from defects and imperfections that may adversely affect the serviceability of the finished product.

3.4 **Repair parts.** The contractor shall furnish repair parts when specified (see 6.1). All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearance in order that such parts may be completely interchangeable and may be replaced or adjusted without requiring modification.

3.5 **Screw threads.** All threaded parts, including screws, bolts, and nuts, shall conform to the applicable requirements of Handbook H28.

3.6 **Nameplate.** Portable pneumatic nut runners shall be furnished with the manufacturer's standard nameplate and, when specified (see 6.1), shall be permanently marked with "U.S." (applicable Department/Agency), and the contract number.

3.7 Lubrication.

3.7.1 *Primary lubrication.* Tools may be provided with either the built-in type or the separate attachment type oil reservoir. The oil reservoir shall be of sufficient capacity so that one filling will supply lubrication to the rotor, cylinder and other moving parts in the path of the air stream for at least 4 hours of continuous operation. An automatic built-in oiling device shall feed the oil from the reservoir during operation of the wrench. The device shall be regulated to prevent continuous discharge of easily visible quantities of lubricant with the exhaust air.

3.7.2 *Secondary lubrication.* Conventional type lubrication fittings or openings or other suitable means shall be provided for introduction of sufficient lubricant for bearings, gears, and other moving parts not lubricated from the oil reservoir. Lubrication shall be required only once in not less than 48 hours of actual operation. No external means of lubrication shall be required for prelubricated sealed bearings.

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3.8 **Air inlet connection.** The air inlet connection shall consist of a well-secured ferrous fitting, threaded with internal NPT threads. The air inlet shall be fitted with an air strainer or screen which shall be effective in retaining solid particles in the compressed air supply. The strainer or screen shall be of such design that it can be readily removed for cleaning.

3.9 **Bearings.** All rotating parts of the portable pneumatic nut runners shall be provided with commercially available rolling contact bearings. The rotor shaft shall be provided with ball bearings. Needle and roller bearings, which are primarily designed for radial loads, shall not be used in a thrust position. Open type bearings shall be suitably housed and adequately sealed to prevent leakage of lubricant and entrance of dirt and dust.

3.10 **General construction.** Tools shall be of rugged construction so as to withstand, without failure, treatment likely to be encountered under general service conditions. Tools shall be self contained, readily accessible for adjustment or replacement of parts. Parts which require lubrication shall be properly enclosed so as to prevent entrance of foreign particles and leakage of lubricant. Working parts exposed to wear shall be of sufficient hardness to withstand the service required. Parts shall be properly finished by machining or grinding to dimensions within limits established by good commercial practice. Similar parts shall be interchangeable and of good fit when assembled with relative parts of the same size, type, and make. Tools shall be reliable and effective when operated on compressed air lines at gage pressures between 80 and 100 pounds per square inch (p.s.i.).

3.11 **Housings.** Housings shall be strong and rugged to prevent indentation under service conditions. When threaded fasteners are utilized to assemble adjacent housing components, through bolting shall be used in preference to tapped holes. Cap screws, if used, shall engage threads in ferrous housings or ferrous inserts in nonferrous housings by an amount not less than the pitch diameter of the fastener thread. Where ferrous inserts are not used in nonferrous housings, cap screws shall engage threads by an amount not less than 1-1/2 times the pitch diameter of the fastener thread. All threaded fasteners shall be secured with suitable locking devices.

3.12 **Motor.** The motor shall be of the rotary type. The rotor of the motor shall consist of one or more elements accurately balanced with the rotor shaft and shall be propelled by compressed air through the attendant action of vanes or blades made of suitable composition.

3.13 **Spindle.** The drive end of the spindle (chuck driver) shall be wear resistant and shall be provided with a suitable means for engaging and holding sockets. The drive end, when fitted with a socket, shall not show excessive runout which may adversely affect operation of the tool. The drive end of the spindle shall conform to ASA Standard B5.38 for the nominal square, male-drive size specified for the applicable wrench (see table I).

3.14 **Operation.** The tools shall be suitable for forward and reverse operation. The rotational direction of the tool shall be controlled by a conveniently located mechanical device which is independent of the throttle and shall be equipped to lock the device in the selected position. The rotational control shall be plainly and permanently marked to indicate direction of rotation. The tools shall be of the stall type, shall conform to the requirements of table I when subjected to the examination and tests of section 4, and shall be similar to figure 1.

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TABLE I.—*Nut runners, pneumatic, portable—weight, dimensions, and performance requirements*

Size of tool	Weight, max.	Overall length, max.	Nominal square male drive (chuck driver)	Air inlet size	No-load speed, min.	No-load air consumption, max.	Stalling torque, min.
Inches	*Pounds*	*Inches*	*Inch*	*NPT*	*r.p.m.*	*c.f.m.*	*Foot-Pounds*
3/8	8	18	1/2	1/4	500	60	20
5/8	12	20	1/2	3/8	450	75	40
3/4	15	22	3/4	1/2	200	100	80
2	65	26	1	1/2	120	180	400

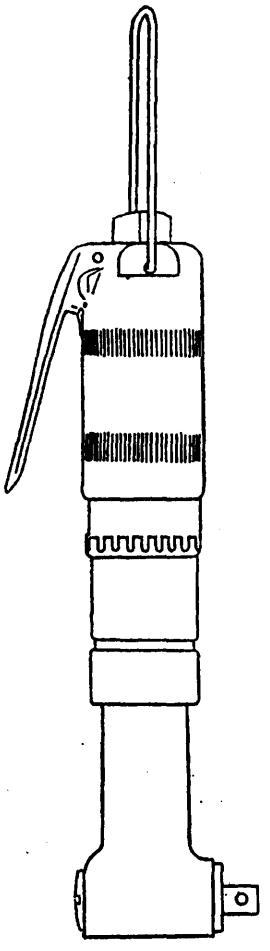


FIGURE 1.—Nut runner, pneumatic, portable.

3.15 **Handle.** Tools shall be equipped with a lever-style handle. The handle shall be fitted with the air-inlet connection.

3.16 **Throttle.** Tools shall be equipped with a lever-style throttle on the handle. The throttle shall be equipped to shut off automatically when released.

3.17 **Instruction book.** One copy of an instruction book consisting of the manufacturer's standard commercial instructions and spare parts list, bound together, shall be furnished with each tool.

3.18 **Identification of product (military use only).** Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.19 **Workmanship.** The tool shall be constructed, assembled, and finished in a thoroughly workmanlike manner to assure procurement of good quality equipment of a neat general appearance. Standards of workmanship shall be such that equipment specified shall comply with the requirements herein.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 . . The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the

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Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Qualification tests.**[1] Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Qualification tests shall consist of the tests specified in 4.6, in addition to the examination of 4.4 and the performance tests of 4.7. Tests shall be conducted on each size of nut runner for which qualification is desired.

[1] Application for qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.2 and 6.3).

4.3 Sampling for acceptance inspection.

4.3.1 *Lot.* All tools of the same size offered for delivery at one time shall be considered a lot for purposes of acceptance inspection.

4.3.2 *Sampling for examination.* A random sample of tools shall be selected from each lot offered for inspection with lot acceptance based on table II.

TABLE II.—*Sampling for examination AQL (approx.)=1.5 percent defective*

Number of tools in a lot	Number of tools in a sample	Acceptance number (defective)	Rejection number (defective)
Up to 8	All	—	—
9 to 15	7	0	1
16 to 25	10	0	1
26 to 65	15	0	1
66 to 180	25	1	2
181 to 500	35	1	2
501 to 1300	50	2	3
1301 and over	75	3	4

4.3.3 *Sampling for operational performance inspection.* A random sample of tools shall be selected from lot offered for inspection with lot acceptance based on table III.

TABLE III.—*Sampling for operational performance inspection AQL (approx.)=1.5 percent defective*

Number of wrenches in an inspection lot	Number of wrenches in sample	Acceptance number (defective)	Rejection number (defective)
Up to 15	3	0	1
16 to 40	5	0	1
41 to 65	7	0	1
66 to 180	10	0	1
181 to 500	15	0	1
501 to 1300	25	1	2
1301 and over	35	1	2

4.4 **Examination.** Each of the sample tools selected in accordance with 4.3.2 shall be examined to verify compliance with this specification. Examination shall be conducted as specified in table IV. Any tools in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective tools in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

TABLE IV.—*Classification of defects in accordance with MIL-STD-105.*

Categories	Defects
Critical:	None defined.
Major:	
101	Size not as specified.
102	Material nonconforming; evidence of unauthorized material used.
103	Tool not readily adjustable or parts not readily replaceable.
104	Lubricated parts enclosure nonconforming.

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TABLE IV.—*Classification of defects in accordance with MIL-STD-105 (cont'd).*

Categories	Defects
Major:	
105	Dimensions of parts not within required limits.
106	Similar parts not dimensionally and functionally interchangeable.
107	Air inlet connection not as specified, damaged or missing.
108	Air strainer missing or damaged, not readily removable.
109	Nameplate nonconforming, illegible, incorrect or missing.
110	Handles not as specified or damaged, not readily removable when required.
111	Throttle nonconforming, not type specified, damaged, inoperable or missing.
112	Rotor nonconforming or damaged.
113	Bearings not as specified, broken, cracked or pitted.
114	Lubrication of tool not as specified; evidence of oil leakage.
Minor:	
201	Packaging and packing not as specified.

4.5 Operational performance inspection. Each sample tool selected in accordance with 4.3.3 shall be subjected to the performance tests of 4.7 to verify compliance with this specification. Any tool in the sample that does not conform to the performance requirements of this specification shall be rejected, and if the number of defective tools in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.6 Test procedures.

4.6.1 *Service test.* Upon satisfactory completion of the examination of 4.4 and the performance tests of 4.7, the tool shall satisfactorily complete 150 hours of actual work under practical service conditions. The test shall determine the ease of operation, general serviceability, suitability, and durability of the tool. Failure during the service test of any components of the tool, exclusive of attachments, shall be cause for rejection.

4.6.1.1 *Examination after service test.* Upon completion of the service test, the tool shall be disassembled and the various parts examined. Any tool having broken, cracked or deformed parts shall be rejected. Tools satisfactorily completing the service test shall again be subjected to the performance tests of 4.7 and shall conform to the applicable requirements of this specification.

4.7 Performance tests.

4.7.1 *Stalling torque and no-load requirements.* Tools shall be subjected to test to determine compliance with the stalling torque, no-load speed and no-load air consumption requirements of table I.

4.7.2 *Performance test conditions.*

4.7.2.1 All performance tests shall be based on operation of the tools with compressed air at a gage pressure of 90 pounds per square inch (p.s.i.), measured at the end of the hose attached to the air meter.

4.7.2.2 A Prony brake or other suitable device shall be used for determining the stalling torque.

4.7.2.3 Air consumption shall be determined with either a liquid displacement meter or a flowmeter of common type.

4.7.2.4 Test tools shall be connected to the air supply by means of a 10-foot length of 5/8-inch inside diameter hose provided at the tool with a coupling conforming to table V.

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TABLE V.—*Coupling requirements*

Air inlet connection of tool (iron pipe size)	Inside diameter of hose coupling
Inch	*Inch*
1/4	0.250
3/8	.375
1/2	.500

4.8 **Inspection of preparation for delivery.** Sample packages and packs shall be selected and inspected in accordance with MIL-P-116 to verify conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

(Civil agencies. Fed. Std. No. 102 should be referred to for definitions of the various levels of packaging protection for supplies and equipment.)

5.1 **Preservation and packaging.**

5.1.1 *Level A.*

5.1.1.1 *Nut runners, pneumatic, portable.* Tools shall be preserved and packaged in accordance with level A requirements of MIL-P-12829. The secondary lubrication system shall be filled to normal operating capacity with the type of lubrication required for normal operation.

5.1.1.2 *Repair parts.* Repair parts shall be preserved and packaged in accordance with level A requirements of MIL-H-775 and MIL-R-196, as applicable. Parts not specifically covered therein shall be unit-protected and packaged in accordance with the guide lines of MIL-P-116. Unless otherwise specified (see 6.1), parts shall be individually packaged except when in sets or quantities greater than one.

5.1.2 *Level C.* Preservation and packaging of tools, except repair parts, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early use.

5.2 **Packing.**

5.2.1 *Levels A and B.* Tools and repair parts, packaged as specified (see 6.1), shall be packed level A and B, as specified (see 6.1), in accordance with the requirements of MIL-P-12829.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking specified in the contract or order (see 6.1), each unit and intermediate package and shipping container shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order (see 6.1), marking for shipment shall be in accordance with MIL-P-12829.

6. NOTES

6.1 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Whether repair parts are required (see 3.4).

(*c*) When special marking on nameplate is required (see 3.6).

(*d*) Size of tool required (see table I).

(*e*) Throttle style required (see 3.16).

(*f*) Applicable levels of preservation, packaging, packing, and marking (see section 5).

6.2 With respect to products requiring qualification, awards will be made only for

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such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL-GGG-N-398 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington 25, D. C., and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3).

6.3 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

6.4 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:
Tools, pneumatic, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

Motor:
Tools, pneumatic, not otherwise indexed.
Truckload minimum weights 21,400; 25,700; and 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

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